Toward Appraisal-Structured Latent Control Representations in Language Models

A Preregistered Hypothesis and Experimental Design for Task-Induced Response-Strategy Modulation

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Abstract

Recent interpretability work shows that instruction-tuned language models contain causally efficacious latent directions for specific strategic behaviors, including refusal, abstention, uncertainty reporting, and evaluation awareness. These results have largely been obtained one behavior at a time. We propose, and preregister a design to test, the hypothesis that several of these effects are special cases of a more general family of *task-induced, appraisal-structured latent control representations*: low-dimensional, potentially correlated representations of the model's own current task situation that partially govern its choice of response strategy across semantic domains. The central methodological contribution is a dissociation, built into a 2×2 factorial design, between *semantic concept tracking* (the prompt names or describes an appraisal-related concept) and *task induction* (the model's own task actually contains ambiguity, incompatible constraints, or low controllability, whether or not it is named). We state four independently falsifiable hypotheses covering task-state dissociation, cross-domain and incremental structure, causal response-strategy modulation, and an exploratory claim about ordered emergence. We specify probing, transfer, and intervention analyses; a battery of competing baselines; and a mandatory lexical-geometry control that rejects the deflationary account in which a direction merely reweights stereotyped output phrases. We characterize the candidate axes as identifiable and selectively manipulable rather than orthogonal, and treat a predictive information-bottleneck reading as optional. The novelty status is a *novel combination*: the components exist separately in prior work; the contribution is their unifying, falsifiable synthesis. This note reports no experimental results and makes no claims about subjective experience, feeling, or related notions.

v0.1.2 pre-data methodological correction. The v0.1.1 family-structure amendment remains binding. Before any data collection, v0.1.2 repairs a non-identifying H1 statistic, replaces it with bidirectional cross-mention transfer inside outer leave-one-domain-out folds, separates H1 from H2 incremental value, fixes training-only layer selection, defines the Layer-0 baseline, strengthens axis-isolation QA, and excludes technical smoke artifacts from scientific analysis. The correction is documented at `preregistration/amendment-v0.1.2.md`. No dataset, model run, activation, generation, intervention, or result exists.

1 Introduction

Interpretability research on large language models has repeatedly found that high-level behaviors are mediated by surprisingly low-dimensional structure in activation space. Representation engineering reads and writes population-level directions associated with concepts such as honesty and harmlessness (Zou et al., 2023); activation engineering steers outputs by adding contrastive direction vectors at inference time (Turner et al., 2023); refusal has been attributed to a single mediating direction (Arditi et al., 2024), and later work argues that refusal involves several geometrically distinct directions that mostly change *how* rather than *whether* a model refuses (Joad et al., 2026). Conditional activation steering makes such interventions context-dependent (Lee et al., 2025a). In parallel, models appear to carry directions for uncertainty and factuality (Teplica et al., 2025), for (un)answerability (Lavi et al., 2026), for evaluation awareness (Nguyen et al., 2025b; Hua et al., 2025), and internal signals that separate recognizing an unsafe input from choosing to refuse it (Han et al., 2025).

These findings share a form: some property of the model's situation is linearly decodable from activations, and intervening on the corresponding direction changes behavior. Yet they have mostly been studied in isolation, one behavior at a time, and often without separating whether the probe tracks the model's *own* task situation or merely tracks *words* describing such a situation.

We propose that a single family of representations may underlie many of these observations. Informally: when a model faces a task that is ambiguous, internally contradictory, low in controllability, or under response pressure, it may form a compact internal representation of that situation, and this representation may help select a higher-level *response strategy*—to answer directly, hedge, ask for clarification, warn, correct a premise, abstain, refuse, or continue conditionally. We call the conjectured family *appraisal-structured latent control representations* (ASCR), borrowing the word "appraisal" from psychology only to name a set of situation variables (relevance, controllability, norm tension, and so on), and not to attribute emotion, feeling, or experience to the model.

The contribution of this note is threefold. First, a precise *dissociation* between concept tracking and task induction, operationalized as a 2×2 factorial design. Second, four preregistered, independently falsifiable hypotheses. Third, an analysis plan with a mandatory control against a lexical-geometry artifact. We report no experiments here; this is a hypothesis and design.

2 Problem Formulation

Let $H_t \in \mathbb{R}^d$ denote the residual-stream activation of an instruction-tuned language model at token position t and a chosen layer. We keep four notions distinct throughout: the *token position* t; the *network layer*; whether H_t is a *prompt-state* activation (at or before the final prompt token) or a *generated-response state* (at an early generated token); and the *design condition* of the prompt.

We hypothesize a small family of control coordinates Z_t extracted from H_t by a projection-like map,

$$Z_t = \Pi_{\mathcal{A}}(H_t), \qquad \dim(Z_t) \ll \dim(H_t). \tag{1}$$

Equation (1) is a *candidate* description, not a proven natural object. We do not assume a single universal $\Pi_{\mathcal{A}}$ shared across models or layers, and we do not fix $\dim(Z_t)$ in advance. Candidate coordinates include signed relevance, uncertainty, goal congruence, controllability/coping capacity, rule/norm tension, and response pressure; we treat goal congruence and controllability/coping as priority hypotheses rather than settled axes.

The **target variable** is not the raw next-token distribution but a higher-level *response strategy* s drawn from a small taxonomy $\mathcal{S}$ (direct compliance, calibrated answer, hedging, clarification request, warning, correction, abstention, refusal, conditional continuation). We use "response strategy" and "policy mode" descriptively; we do not claim that an autoregressive model is an external reinforcement-learning agent.

The core distinction is between two ways an appraisal-related property can be present in a prompt:

Concept tracking.
The prompt *names or describes* the property (e.g. the words "uncertain", "conflict", "out of my control"), while the model's own task remains straightforward.

Task induction.
The model's *own current task* contains the property (missing information, incompatible instructions, no permissible resolving action), whether or not the prompt names it.

The central question is whether the model internally represents its *own* current task situation in a form that helps determine its response strategy, rather than merely representing appraisal-related words or third-party described situations.

3 Related Work

Reading and steering latent directions. Representation engineering places population-level representations at the center of analysis and manipulates high-level attributes (Zou et al., 2023). Activation addition computes steering vectors from contrastive prompt pairs and adds them at inference time (Turner et al., 2023). Unsupervised methods elicit latent behaviors without labeled directions (Mack and Turner, 2024). These establish that low-dimensional directions can causally move behavior—a premise we adopt rather than re-establish.

Refusal and conditional control. Arditi et al. (2024) attribute refusal to a single mediating direction across many open models; Joad et al. (2026) find multiple category-specific refusal directions that primarily change the manner of refusal. Conditional activation steering gates such interventions on input context (Lee et al., 2025a). These are our closest neighbors on the "strategic behavior" axis, but they target one behavior family (refusal) rather than a general situation representation.

Uncertainty, answerability, and evaluation awareness. SCIURus locates uncertainty in the same circuitry responsible for factuality (Teplica et al., 2025); linear directions detect (un)answerability and control abstention (Lavi et al., 2026), and hidden states encode query answerability early in decoding (Slobodkin et al., 2023). Abstention itself decomposes into distinct axes of answer correctness and question answerability (Wagner, 2026), and a functional dissociation via sparse autoencoders separates uncertainty features from correctness features (Patel et al., 2026). Models linearly represent whether they are being evaluated (Nguyen et al., 2025b), and this state can be steered so a model acts "deployed" (Hua et al., 2025). Internal-activation safety signals separate unsafe-input recognition from the refusal decision (Han et al., 2025). Several of these are genuinely *task-induced* states, and they motivate the conjecture that a broader family exists; they also narrow the novelty of any *single* ASCR axis.

Task representations and affect geometry. Task representations form just-in-time and transfer across instances (Li et al., 2025). In the affective domain, appraisal variables are causally implicated in *third-person* emotion inference (Tak et al., 2025); emotion-concept representations track an operative concept and causally influence behavior (Sofroniew et al., 2026); and a valence/arousal subspace with circular geometry affords multi-behavioral control (Sun et al., 2026). This work supplies the multi-axis, appraisal-structured vocabulary we borrow, but it targets emotion concepts or affective generation rather than the model's own response-strategy selection.

Norm/role tension, difficulty, and multi-attribute steering. Task-induced norm and role conflicts are separately decodable and steerable: contextual privacy norms decompose into steerable contextual-integrity parameters (Wang et al., 2026), and system-versus-user role conflicts are linearly encoded and manipulable (Zeng, 2025), both close to the norm-tension axis. Generic task difficulty is itself a linear, decodable property (Lee et al., 2025b), which we therefore treat as a mandatory baseline rather than as appraisal structure. Most directly, multi-attribute steering learns several attribute directions jointly while enforcing near-orthogonality to reduce interference (Nguyen et al., 2025a). These works establish the individual axes and the feasibility of multi-attribute control, but they do not test whether the axes form a *shared task-induced control family*; and in contrast to their orthogonality constraints, we frame the axes as identifiable and selectively manipulable rather than necessarily orthogonal.

Privileged, verbalizable representations. A separate line of work reports that language models maintain a small, privileged set of internal representations that are available for report and internal reasoning atop a much larger volume of automatic processing—a "workspace"-like "J-space" identified by surfacing the concepts a model is poised to verbalize (Gurnee et al., 2026). We note this only cautiously and with a clear boundary: such a verbalizable readout is at most a *possible privileged readout or broadcast substrate*, and we do not treat it as identical to the appraisal-structured control representations conjectured here. Whether any task-induced control coordinate is (or is not) verbalizable in that sense is an open, separable empirical question; ASCR is defined by its role in response-strategy selection, not by verbalizability.

Position of this work. To our knowledge, we found no prior work that *jointly* tests (i) a task-induced, system-relative state dissociated by design from concept tracking, (ii) an integrated multi-axis appraisal structure governing response strategy, (iii) cross-domain transfer of that structure, (iv) a mandatory lexical-geometry control, and (v) an explicit test that the axes form a *shared* control family rather than several independent known directions. The components are established separately; the proposed contribution is their unifying, falsifiable synthesis. As sharpened in the v0.1.1 pre-data amendment, demonstrating several individual task-state directions is not itself evidence of a shared family, so the family claim carries its own preregistered test (Section 5 and the amendment). A companion prior-art matrix in the repository records this component-by-component.

4 Appraisal-Structured Latent Control Representations

We now state the hypothesis object more carefully. Consider a distribution over tasks. For a task instance, let $A \in \mathbb{R}^k$ collect the candidate appraisal coordinates (relevance, uncertainty, goal congruence, controllability, norm tension, response pressure). The ASCR conjecture has two parts: that a low-dimensional image of A is recoverable from activations as in Equation (1), and that this image conditions the response strategy.

An optional compression reading. One *optional* formal characterization is that Z_t approximates a predictive information bottleneck: a compression of the activation state that discards task-irrelevant detail while retaining what predicts the appropriate strategy. Writing Y for the strategy-relevant target and $I(\cdot;\cdot)$ for mutual information, a candidate objective is

$$\min_{p(Z_t|H_t)} \; I(H_t; Z_t) \;-\; \beta\, I(Z_t; Y). \tag{2}$$

Equation (2) is a *candidate compression principle*, not a demonstrated mechanism, and we do not claim that Transformers have been shown to implement it. The core hypothesis is designed to survive even if this bottleneck reading is wrong; nothing in the empirical predictions depends on Equation (2).

Identifiability, not orthogonality. We do *not* assume the coordinates are mutually independent, structurally uncorrelated, or geometrically orthogonal. We instead ask whether they are *identifiable*: separately decodable, cross-domain transferable, interventionally identifiable, incrementally predictive, and selectively manipulable. Correlations among axes are measured, not assumed away. We summarize the conditioning claim as a family of per-axis coordinates that are only approximately separable,

$$Z_t = (z_t^{(1)}, \ldots, z_t^{(k)}), \qquad \mathrm{Cov}(z_t^{(i)}, z_t^{(j)}) \neq 0 \text{ permitted}, \tag{3}$$

and the response-strategy claim as a conditioning of the strategy on these coordinates and the context c_t,

$$p(s \mid \text{context}) \;\approx\; g(Z_t,\, c_t)\,, \tag{4}$$

where g maps control coordinates and residual context to a distribution over strategies $s \in \mathcal{S}$. Equations (1), (2), (3), and (4) are the four core equations; all other relations are kept in prose.

5 Hypotheses

We preregister four hypotheses; H1–H3 are load-bearing and H4 is exploratory.

H1 — Task-state dissociation. Actual task-induced uncertainty, constraint conflict, or low controllability is decodable from activations independently of the presence of appraisal-related vocabulary; a probe tracks the real task condition more strongly than lexical concept mentions. The historical v0.1.1 difference between task-state-probe and concept-probe accuracy on cells B/C is non-identifying: on B/C the labels are exact complements, so two perfect probes and two chance probes can both yield a zero accuracy difference. It is therefore a non-deciding diagnostic after v0.1.2.

The replacement H1 feasibility analysis fits task state on A/C in three domains and evaluates on B/D in the held-out domain, then reverses the transfer (B/D to A/C), for every outer domain. Whole matched groups remain on one side of every boundary. Layer and logistic regularization are selected only with inner matched-group folds from the outer-training domains. The primary statistic is balanced accuracy from pooled out-of-fold predictions across domains and both directions, with a cluster bootstrap over matched groups. A lower 95% confidence limit above 0.5 is positive feasibility; an upper limit at or below 0.5 weakens H1 if all technical and QA gates pass; overlap is indeterminate. Both directions must support transfer before bidirectional robustness is claimed.

H2 — Cross-domain and incremental structure. Task-induced control representations transfer across semantically disjoint domains and predict response strategy beyond lexical features, prompt length, surface sentiment, valence/arousal baselines, generic difficulty, confidence or unanswerability alone, and a single refusal direction. H2 is not merged with H1: under the identical double-crossed groups and folds, its primary comparison is hidden state versus one author-approved frozen prompt-embedding model, using paired held-out log-loss. The embedding receives canonical user-visible prompt text, not chat-template artifacts. Its identity and immutable revision remain unresolved pre-data blockers rather than post-data choices.

H3 — Causal response-strategy modulation. Intervening along an identified task-state direction shifts response strategy in the predicted direction. Both sufficiency (matched-norm activation addition) and necessity (directional ablation or counter-steering) are tested, and the intervention preserves general linguistic competence as far as possible.

H4 — Ordered emergence (secondary). Coarse signed-relevance or task-pressure signals may become decodable earlier than integrated task-appraisal structure, while explicit response-strategy commitment may appear later or closer to generation. This is labeled secondary because the literature does not justify a strong confirmatory ordering claim.

Family-structure test (v0.1.1 amendment). Succeeding on H1–H3 for several axes is compatible with several independent, already-known directions and does not, by itself, establish a shared appraisal-structured control *family*. The v0.1.1 pre-data amendment therefore preregisters a separate family-structure test with three components that must *all* hold, i.e. the family claim requires $A \wedge B \wedge C$: (A) a shared low-rank model beats equally complex separate per-axis models on pooled held-out log-loss under nested cross-validation (rank grid restricted so that three axis directions cannot trivially count as a family); (B) structured behavioral specificity, where each axis moves a preregistered target set of response strategies rather than producing a generic refusal, negativity, or fluency-degradation effect; and (C) an incremental shared contribution to held-out response-strategy prediction beyond known single directions and simple baselines. The full decision rules are in the preregistration (`preregistration/analysis-plan.md` and the amendment).

6 Preregistered Experimental Design

Core 2×2 design. For each candidate axis, and holding the broad semantic task fixed, we construct four conditions crossing the actual task state with the appraisal-concept mention:

Actual task state	Concept mentioned	Interpretation
absent	absent	neutral control
absent	present	concept tracking only
present	absent	pure task induction
present	present	combined condition

Prompts are authored in *matched groups*: one base task instantiated across all four cells, differing only along the two intended factors, sharing a matched group identifier. Whole groups are assigned to an outer training or test domain; cell restrictions choose used siblings without moving unused siblings across that boundary. The primary H1 analysis asks whether task-state decoding transfers when the concept-mention level and semantic domain both change.

Axes and domains. Primary pilot axes are uncertainty/unanswerability, rule/instruction conflict (norm tension), and controllability/coping; signed relevance and goal congruence are exploratory. We use at least four semantically distinct domains—software debugging, scheduling and planning, policy-constrained assistance, and document editing / creative problem solving—so that no single sensitive topic dominates. The design supports leave-one-domain-out transfer.

Pilot size. Mini-0 targets at least forty complete uncertainty-axis matched groups, distributed as evenly as feasible across the four domains, solely for pipeline and feasibility assessment. The broader historical sketch of roughly twenty-four matched groups per axis and a few hundred reviewed prompts is not a powered target. No formal power analysis has been performed. Effect-size and variance estimates from a valid feasibility run, if later produced, must inform a separately versioned pre-data power/precision plan before any confirmatory collection.

Model and activations. The primary model is an open-weight, instruction-tuned model in the $\sim$7–9B range (default: `Qwen/Qwen2.5-7B-Instruct`); the exact immutable revision hash is recorded before data generation. An optional replication model may be named for the scaling phase, which does not imply that replication has occurred. We record residual-stream activations at the final prompt-token position and a small preregistered set of early generated-token positions, across all layers or a preregistered subset, labeling each activation with token position, layer, prompt-vs-generated flag, and design cell.

A tokenizer-only inspection on disposable strings, which loaded no model weights, showed that the inspected Qwen assistant-generation template ended in an identical newline token across prompts. Hidden layers retain the final non-padding prompt token (greatest index with attention mask one). At Layer 0, that final-token embedding is only a sanity check; the informative baseline mean-pools only the user-visible content span, excluding padding, system/role/template spans, assistant prefix, and special tokens. The binding tokenizer revision remains unfrozen and blocks a run.

Probes and transfer. We use interpretable linear methods first—logistic regression, ridge regression, and linear discriminant analysis—with matched-group train/validation/test splits, leave-one-domain-out evaluation, multiple random seeds, bootstrap confidence intervals, and separately declared H1, H2, H3, and family-test comparison families. The single pooled H1 and H2 primary statistics are not FDR-corrected merely because secondary families exist. High in-domain probe accuracy is *not* treated as evidence of a general state; the load-bearing result is cross-domain transfer and incremental predictive value.

Run-readiness and artifact separation. Run-ready QA must confirm each registered task-state and concept-mention value and the absence of the two non-target primary axes. The current configuration remains blocked by unresolved model, tokenizer, prompt-embedding, and layer-grid choices. A technical smoke manifest must use disposable prompts and declare itself ineligible for scientific analysis; it is structurally incompatible with a scientific Mini-0 manifest and cannot enter selection, variance, effect-size, or power calculations.

Causal interventions. We preregister activation addition (sufficiency) and directional ablation / counter-steering (necessity), each with matched-norm random-direction and PCA-direction controls, several intervention strengths, several layers, and multiple seeds. Effects should be dose-dependent where predicted, strategy-specific, reproducible, and not reducible to degraded fluency or a generic refusal shift. We separately track changes in wording, style, semantic content, and high-level strategy, and monitor competence on off-target neutral tasks.

7 Baselines and Alternative Explanations

The task-state representation must add incremental value over each of the following, individually and jointly: bag-of-words/lexical classifiers, sentence or prompt embeddings, prompt length, token entropy, independently rated difficulty, sentiment/signed-valence, a valence/arousal subspace, a confidence/unanswerability direction, a single refusal direction, an evaluation-awareness direction, and random and PCA subspaces of matched dimensionality. Support for ASCR requires beating the *strongest* of these, not merely the weakest.

The primary H2 comparator is one frozen prompt-embedding model on canonical prompt text. No model has yet been author-approved, so its immutable revision, license, pooling, and input rule remain run blockers. TF–IDF is diagnostic under cross-mention transfer and remains secondary under ordinary LODO; predictable vocabulary-shift failure cannot serve as the primary H2 comparator.

Mandatory lexical-geometry control. A deflationary account holds that a "control direction" only changes the probability of stereotyped output phrases ("I can't", "sorry", "however", "I am uncertain", standard refusal templates). We therefore require: paraphrased output strategies; meaning-based strategy labels; removal or masking of stereotyped lexical markers during evaluation; alternative phrasings expressing the same strategy; analyses of whether behavior changes persist after lexical normalization; and matched prompts whose vocabulary differs while task structure is held constant. *A direction that changes stereotyped words without changing the underlying response strategy does not support the control-state hypothesis.*

8 Falsification Criteria

The central interpretation is weakened or rejected if any of the following hold: probes primarily track appraisal-related words rather than real task state; cross-domain transfer fails; lexical or simple embedding baselines perform equally well; valence/arousal, confidence, difficulty, or refusal alone explains the result; causal intervention changes only stereotyped wording; steering effects are indistinguishable from matched random controls; ablation has no specific behavioral effect; intervention destroys generic competence; candidate axes cannot be separately manipulated; controllability/coping adds no value beyond simpler baselines; results fail to replicate across seeds; or response-strategy labels are unstable or evaluator-dependent. A negative result is reported plainly as such.

9 Predictions for Existing Latent Representations

The following are cautious, testable predictions, not results:

1. Published emotion-concept vectors may exhibit a lower-dimensional factor structure partly aligned with appraisal-related variables.

2. A valence/arousal subspace may explain some variance yet fail to distinguish response strategies that differ primarily in controllability, coping, or norm tension.

3. Fear-like withdrawal and anger-like correction offer an intuitive motivating contrast, but the main claim is stated in non-emotional, strategic terms (abstention vs correction).
4. Safety systems that separate unsafe-input recognition from the refusal choice may be a partial instance of an appraisal-versus-action separation.
5. Multi-category refusal directions may be a restricted special case of a broader family of situational control axes.
6. Task-induced state probes should transfer better across domains than probes based only on semantic concept mentions.

10 Limitations and Non-Claims

This note reports no experimental results and establishes no mechanism. It does *not* claim, and its design cannot establish, subjective experience, feeling, consciousness, sentience, suffering, phenomenal states, biological equivalence, a contemplative or Buddhist mechanism inside a Transformer, a universal architecture shared by all language models, or the final correct number of latent dimensions. The words "appraisal" and "control" are used technically: "appraisal" names a set of situation variables, and "control" names an influence on response-strategy selection, with no anthropomorphic or agentive commitment. Findings, if obtained, would be specific to the tested models, axes, and domains. The information-bottleneck reading of Section 4 is optional and not relied upon.

11 Research Roadmap

The immediate step is authoring and validating the matched-group pilot corpus and freezing the model revision. Next is the probing and transfer analysis under Section 6, followed by causal interventions with the full control battery. Only if the primary hypotheses survive the controls do we proceed to a powered confirmatory study with a pre-study power analysis, the exploratory axes, an additional held-out domain, and a replication model. Data, prompts, labels, seeds, and the frozen model revision hash will be released to make any results reconstructable.

12 Conclusion

We have set out, and preregistered a design to test, the hypothesis that a family of task-induced, appraisal-structured latent control representations of a model's own task situation partially governs its response-strategy selection, and that several existing single-behavior findings may be special cases of this family. The distinctive commitments are a built-in dissociation of task induction from concept tracking, an identifiability (rather than orthogonality) framing of the axes, and a mandatory lexical-geometry control. The novelty status is an honest *novel combination*: the components exist in prior work; the contribution is their unifying, falsifiable synthesis. Whether the hypothesis survives its own falsification criteria is an empirical question this note is designed to make answerable.

Data and Code Availability

This is a design note; no data or results accompany it. The design scaffold, preregistration documents, prior-art matrix, and citation-verification log are in the accompanying repository. This unreleased v0.1.2 correction has no version-specific DOI. Historical archives remain unchanged: v0.1.1 is DOI `10.5281/zenodo.21335529`, v0.1.0 is `10.5281/zenodo.21294933`, and the concept DOI is `10.5281/zenodo.21294932`.

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